Population-Based Training

Six senses of one phrase, and what happens to the population when training stops

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The phrase reaches my reading already attached to something else. The PSRO survey closes its framework section by saying PSRO can be classified as a variant of population-based training, and the label sticks, so every PSRO paper I read inherits a name coined for a different algorithm solving a different problem. The two are not the same thing and the difference is not cosmetic. In one of them the population is scaffolding that gets thrown away. In the other the population is the answer.

This file separates them. It enumerates where the word appears in my own files, sets those uses against the senses the field carries, assigns each occurrence, and leaves the two occurrences that will not sit still marked as ambiguous.

Contents

1 The occurrences

Seven, across three projects. None of them is the hyperparameter sense.

the user population, the attacker population

> *"Emulated: the user population (GHOSTS personas, scripted), the attacker population (two fixed-class profiles, scripted)"*
>
> FOE-Dreamer, `sections/04_threat_model_and_testbed.tex`, on what operational means here

The website page states the same split in one clause.

> *"The user and attacker populations and the CVE catalogue are emulated."*
>
> website, FOE-Dreamer, on what operational means here

strategy population

> *"assembling a strategy population that represents the full game under a computational budget"*
>
> website, AcceleratePSRO, on the strategy exploration problem

which strategies enter

> *"which strategies enter the empirical game governs what the analysis can deliver"*
>
> website, Learn Structure, on the budget argument

restricting the strategy space

> *"Are we allowed to limit the strategy space by saying there are clusters and some strategies are infeasible, or some other justifications, and exclude strategies from the strategic spaces as we like, as long as we can find an excuse?"*
>
> Learn Structure, `Notes/Latex/sections/notes_0630.tex`

The answer written against it, and the part of it that stays open:

> *"Reading (a) is correct: restricting the strategy space is standard, not cheating."*
>
> Learn Structure, `cards/GenerateMDP-09-strategies-e-restrict.tex`

> *"Restriction error: the chosen subset vs the full policy space. NOT bounded."*
>
> same card

population management

The Foundation Model Self-Play entry in the PickYourBattles status notes records what I want out of that line of work: embedding-based novelty and dimensionless MAP-Elites, imported as ways to manage the PSRO strategy set rather than as a training scheme of their own.

population exploitability

Li et al. 2022, recorded in the same notes, tie the response oracle to the quality of the equilibrium approximation through population exploitability, which reflects the coverage of the policy hull, the convex combination of strategies in the restricted game. Their result is that a larger hull implies lower population exploitability. Here the population is neither scaffolding nor environment. It is the thing being measured.

2 The senses

2.1 Sense one, the hyperparameter population

Jaderberg et al. 2017, the paper the phrase was coined in.

> *"a simple asynchronous optimisation algorithm which effectively utilises a fixed computational budget to jointly optimise a population of models and their hyperparameters to maximise performance"*
>
> Jaderberg et al. 2017, abstract

> *"PBT discovers a schedule of hyperparameter settings rather than following the generally suboptimal strategy of trying to find a single fixed set to use for the whole course of training"*
>
> same

Members are ranked, the worst are overwritten by mutated copies of the best, and the run ends with one model and one hyperparameter schedule. The population exists so that a schedule can be discovered. It does not survive the run and no result is stated over it.

2.2 Sense two, the strategy population

Bighashdel et al. 2024, the PSRO survey, Section 2.3, titled RL View of PSRO: Population-Based Training.

> *"In PSRO, a restricted game maintains a population of strategies (also known as policies, i.e., RL agents) and expands this population by introducing new strategies that respond to a specific distribution (mixture) of strategies within the existing population."*
>
> Bighashdel et al. 2024, Section 2.3

> *"Consequently, PSRO can be classified as a variant of population-based training."*
>
> same

The set is the analysis object, and the cost of building it is the research problem:

> *"This challenge of restricted game construction with minimum computational cost (i.e., with the fewest strategies required) is described as the strategy exploration problem"*
>
> Bighashdel et al. 2024, Section 2.2

Every member is kept, because dropping one changes the restricted game and therefore changes the equilibrium and the regret computed over it. The population is what the answer is stated about.

2.3 Sense three, co-evolving species

From the same survey's account of where PSRO came from.

> *"Co-evolutionary methods evolve multiple populations of (different) species in parallel. In this setting, there is no explicit fitness function, but the fitness of a population member depends on how well it interacts with members of other populations."*
>
> Bighashdel et al. 2024, Section 2

Plural populations, one per species, and fitness defined only relationally. This is the sense the diversity measures descend from, and it is why behavioral diversity and response diversity are two measures rather than one.

2.4 Sense four, the opponent set

Czempin and Gleave 2022 use the phrase for something that is neither of the first two.

> *"We analyze a defense using population based training to pit the victim against a diverse set of opponents."*
>
> Czempin and Gleave 2022, abstract

One learner, the victim, is trained. The population is the distribution it trains against, and the paper reports that how hard the victim is to exploit tracks the size of that opponent set. No hyperparameters are searched and no restricted game is solved.

2.5 Sense five, the league

AlphaStar (Vinyals et al. 2019) runs a population with an internal division of labor: main agents, main exploiters trained only against the current main agent, and league exploiters trained against the league. Membership is structured by role, matchmaking is by a performance score, and members are added and reset on schedule. This is sense two with a fixed organizational chart imposed on the set, and it is the version most cyber papers reach for when they cite population training.

2.6 Sense six, the sampled population

In epidemiology and in human-subjects work, population-based names a sample drawn to stand for a defined population rather than for whoever arrived at the clinic. The property claimed is representativeness, the failure mode is selection bias, and nothing is trained at all. This is the sense a collaborator from the behavioral side will hear first, and it is the one my own testbed sentence is closest to.

3 The assignment

Each occurrence against the sense it belongs to. Two are marked ambiguous rather than resolved.

Occurrence	Sense	Why
user population, attacker population	ambiguous, six or two	Scripted, fixed, standing in for real users and real attackers, which is sense six. Two attacker profiles are also a strategy set of size two that never expands, which is sense two frozen at initialisation.
strategy population under budget	two	The set represents the full game, and the budget is spent on choosing its members.
which strategies enter the empirical game	two	States the same claim as a consequence for the analysis.
restricting the strategy space	two	The question is about which members are admitted, and the answer names the unbounded error that admission decision leaves.
population management, novelty, MAP-Elites	three into two	Mechanisms from the evolutionary line, imported to decide what enters a restricted game.
population exploitability	two	Regret of the mixture over the policy hull, a quantity defined only over a set.
PSRO as a variant of population-based training	ambiguous, two labeled one	The mechanism described is sense two. The name attached to it is sense one.

The census result worth stating: sense one appears nowhere in my own writing. I have never run a hyperparameter population and never will in this line of work. The phrase enters my vocabulary only as a label the survey hands to something else.

4 Where the senses disagree

> **The label is borrowed from an algorithm whose loop is not the one being run**
>
> Jaderberg's population is discarded. The run produces one model and one schedule, and the members were a search device.
>
> The PSRO population is retained in full. Every member changes the restricted game, and the equilibrium and the regret are defined over the set.
>
> The same three words therefore name two opposite fates for the population. Any sentence that says a method inherits guarantees or intuitions from population-based training needs to say which one, because nothing carries across.

> **Two populations in one testbed, with nothing in common**
>
> Daedalus has an attacker population of two scripted profiles. It is fixed for the life of the experiment, it is part of the environment, and no gradient touches it.
>
> AcceleratePSRO has a strategy population that expands every iteration. It is the output of the

> algorithm and the object the result is stated over.
> The word is the same in both sentences and the referent shares no property except plurality.
> Whichever page a reader arrives from, the other sentence will mislead.

> **Diversity is measured against two different things**
>
> In sense one, diversity is spread in hyperparameter space, and its value is that the search does not collapse early.
> In sense two, Liu et al. 2021 split it: behavioral diversity is the difference between action-state coverages, response diversity is the distance between the payoff vector a new strategy induces and the payoff vectors already in the restricted game. One is a property of how the strategies behave, the other of what they do to the game.
> A population can score high on behavioral diversity and add nothing to the restricted game, which is the failure the response measure exists to catch. Importing a diversity mechanism from sense one gives no reason to expect the second kind.

> **A paper title using the phrase for neither of the two main senses**
>
> Czempin and Gleave, *Reducing Exploitability with Population Based Training*, train one victim against a set of opponents. No hyperparameter search, no restricted game, no equilibrium.
> The title reads as sense one to a machine-learning audience and as sense two to a game-theory audience, and the contents are sense four. This is the concrete reason the entry has an operational test rather than a definition alone.

5 Placement on the five dimensions

Using the axes from the dictionary overview page.

- **Object.** Sense one, configurations of a single learner. Sense two, strategies of a player. Sense three, species. Sense four, opponents. Sense five, agents holding assigned roles. Sense six, actors in the world the study is about.
- **Epistemic operation.** Optimize, analyze, evolve, harden, rank, sample, in that order.
- **Specificity.** Sense one is fixed to one task. Sense two is fixed to one game. Sense six is fixed to one defined class of real people, and its whole claim is about that class rather than about the sample.
- **Purpose.** One better model, then an equilibrium and a regret number, then a learner harder to exploit, then a result that generalizes past the sample.
- **Fidelity.** Senses one to five are abstract representations. Sense six is executable in my case, because the GHOSTS personas actually run on the testbed and generate traffic.

Two of these axes move together and the table cannot show it. The more a population stands for particular real actors, the further it moves toward executable reproduction, because standing for real actors is a claim you can only support by producing their behavior. Specificity and fidelity rise as one along the path from sense one to sense six, and a scheme that reads the placement as an axis-aligned box will admit a combination that does not exist: a maximally specific population held at a maximally abstract fidelity.

6 The entry

Definition. A training scheme that holds more than one learner at once and operates on the set rather than on any single member, so that what is selected, measured, or reported is a property of the set.

Distinguishes it from. Self-play, where a strategy trains against its own current copy, so the set has size one at every moment and there is no diversity to select over. The survey draws exactly this contrast in Section 2.3 before applying the label. It also differs from ensembling, where several models are trained independently and combined only at inference, and never compete for training resources against each other.

Operational test. Ask what survives training.

1. If the members are discarded and one model ships, the population was a search device and the phrase is Jaderberg's.
2. If every member is kept because the result is stated over the set, the phrase is the PSRO one.
3. If the members were never trained at all, the word is describing the environment and the word training does not apply to them.

A single question settles all three: name the object the paper's headline number is computed over. A score for one model gives case one. A regret or an exploitability gives case two. A description of the setting gives case three.

In my project. AcceleratePSRO takes sense two as its subject: assembling a strategy population that represents the full game under a computational budget, with the acceleration levers acting on the loop that builds it. Learn Structure supplies the admission question, which strategies enter the empirical game, and the finding that the restriction error is not bounded. FOE-Dreamer uses the word in sense six, for the scripted user and attacker populations of Daedalus, and that use has nothing to do with a training scheme. When the three tracks are written about together, the word needs qualifying every time it appears.

7 Sources

- Bighashdel, A., Wang, Y., McAleer, S., Savani, R., and Oliehoek, F. Policy Space Response Oracles: A Survey. *Proceedings of IJCAI 2024*, Survey Track, pages 7951 to 7961. Sections 2, 2.2 and 2.3 read from the proceedings PDF. The sections B.1.1 through B.1.8 extraction of the same survey in `PickYourBattles/Notes/Latex/project status.md` supplied the entries on diversity measures and on population exploitability.
- Jaderberg, M., Dalibard, V., Osindero, S., Czarnecki, W. M., Donahue, J., Razavi, A., Vinyals, O., Green, T., Dunning, I., Simonyan, K., Fernando, C., and Kavukcuoglu, K. Population Based Training of Neural Networks. arXiv 1711.09846, 2017.

- Czempin, P., and Gleave, A. Reducing Exploitability with Population Based Training. arXiv 2208.05083, 2022.
- Vinyals, O., et al. Grandmaster level in StarCraft II using multi-agent reinforcement learning. *Nature* 575, 2019, for the league and its three agent roles.
- Liu, X., et al. 2021, for behavioral diversity and response diversity, and Li, W., et al. 2022, for population exploitability and the policy hull, both as stated in the survey and in the status notes.
- My own files: the AcceleratePSRO and Learn Structure website pages, the FOE-Dreamer threat model and testbed section, and, in the Learn Structure notes, `sections/notes_0630.tex` together with the card `GenerateMDP-09-strategies-e-restrict`.